NIEK WELS

MSc Geo Information Science



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Wageningen, Netherlands

1 April 2004

Dutch (native)
English (C2) & German (B1)

PROFILE

Curious and analytical person with a passion for untangling complex societal and spatial issues. As a Human Geography graduate, I combine this with my interest in how people and environments are connected. My drive lies in visualising and simplifying complicated problems, in particular to be able to quantify complex problems better and more efficiently with AI. I am looking for a role in which I can use my analytical skills and my ability to connect with people to make abstract information accessible, and in doing so contribute to practical solutions for the world of tomorrow.

WORK EXPERIENCE

2020 – present
Albert Heijn
Service Employee
- Problem-solving attitude under pressure
- Customer-focused communication skills

September 2024 – February 2025
Municipality of Tilburg, Stadhuisplein 130, Tilburg
Intern Neighbourhood Manager, Tilburg Noord
- Safeguarding the long-term planning of developments
- Communication to stakeholders

EDUCATION

2025 – present
Wageningen University & Research
- Master: Geographic Information Science
  - Specialisation: AI (deep learning & foundation models)

2022 – 2025
Utrecht University
- Bachelor: Human Geography and Spatial Planning
  - Track: Economic Geography
  - Minor: European Governance

2016 – 2022
Van Maerlantlyceum, Eindhoven
- VWO (pre-university education)
  - Profile: Culture & Society and Economics & Society

TECHNICAL KNOWLEDGE

- Python / R
  - scikit-learn, PyTorch, GDAL, rasterio, geopandas, shapely, terra, stars & sf
- Remote sensing
  - Sentinel-1 / 2, aerial imagery & LiDAR
- Machine learning
  - Regression / classification tasks
- Deep learning
  - Geospatial foundation models
  - Graph Convolutional Neural Network
- Git
  - Version control

SKILLS

- Adaptive
- Eager to learn
- Learns quickly and easily
- People-oriented
- Socially skilled
- Stress-resistant

PROJECTS

- Master Thesis: “Can geospatial foundation models reduce annotation costs for sample-based area estimation for land cover classification change?”
  - Used geospatial foundation models to make quantifying landscape change more efficient
- ACT project (CGI & Kadaster)
  - Investigated the potential of deep learning for aggregating very detailed maps (BGT) to smaller scales (BRT).
- Click here for further explanation and more projects

INTERESTS

- Cooking and culinary discoveries, mainly in Italian cuisine
- Music festivals
- Road cycling